

BRIGHT EMERALD COMPUTER ACADEMY

Empowering youth with Practical Digital Skills



MICROSOFT POWER BI FOR BEGINNERS

A Visual, Interface-Led Introduction to Data Analysis and Dashboarding

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• Complete Lesson Guide • Course Outline • Description & Detailed Training Notes

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1. Course Description

This course, Microsoft Power BI for Beginners, is a hands-on, visually driven introduction to Power BI Microsoft's leading business intelligence and data visualization platform. Designed for complete beginners, the course places special emphasis on illustration and interface familiarity: learners begin by seeing and understanding exactly where every tool lives on screen before they are asked to use it. By the end of the course, learners will be able to confidently open Power BI Desktop, import and clean data, build a data model, design interactive visuals, and publish a finished report to the Power BI Service.

1.1 Who this Course is for

- Absolute beginners with little or no prior experience in data analytics or business intelligence tools.
- Students, administrators, NGO/M&E officers, and small-business owners who need to turn spreadsheets into visual reports.
- Excel users who want to move beyond static tables and pivot charts into interactive dashboards.
- Anyone preparing for entry-level data analyst, reporting, or MIS/M&E roles.

1.2 Prerequisites

- Basic computer literacy (using a mouse/keyboard, opening and saving files).
- Basic familiarity with Microsoft Excel (rows, columns, simple formulas) is helpful but not mandatory.
- A Windows PC (Power BI Desktop is a free Windows application) with at least 4 GB RAM recommended.

1.3 Course Duration

16 hours total, delivered across 10 modules. Recommended pace: 2 sessions per week of 4 hours each, combining short demonstrations, guided practice, and independent exercises.

1.4 Learning Objectives

By the end of this course, learners will be able to:

1. Explain what Power BI is and how it fits into the wider Microsoft data ecosystem.
2. Confidently navigate the Power BI Desktop interface Ribbon, Views, Canvas, and Panes.
3. Connect to common data sources such as Excel, CSV files, and web/database sources.
4. Clean and reshape raw data using the Power Query Editor.
5. Build a simple data model and understand table relationships.
6. Write basic DAX formulas to create measures and calculated columns.
7. Design clear, interactive visualizations and dashboards.
8. Publish and share reports through the Power BI Service.

1.5 Teaching Methodology

- Live, on-screen demonstration for every new tool or pane, supported by labelled illustrations.
- **"Watch, then do"** practice: learners repeat every demonstrated action on their own machine immediately.
- Short quizzes and hands-on exercises at the end of every module.
- A capstone project in the final module that ties every skill together.

1.6 Assessment

- Formative: in-class exercises and module quizzes (40%).
- Summative: final capstone dashboard project and short oral walkthrough (60%).
- A Certificate of Completion is issued by Bright Emerald Computer Academy to learners who complete the capstone project.

2. Course Outline

The table below summarizes all ten modules, their key skills, and estimated duration. Full teaching notes for every module follow in Section 3.

Module	Topic	Key Skills Covered	Duration
1	Introduction to Business Intelligence & Power BI	What is BI; what is Power BI; the 3 building blocks (Desktop, Service, Mobile)	2 hours
2	Installing Power BI Desktop & Touring the Interface	Installation; Ribbon, Canvas, Panes, Views (illustrated)	
3	Connecting to Data Sources	Get Data; Excel, CSV, Web, Databases; Navigator	2 hours
4	Cleaning Data with Power Query Editor	Remove errors, change types, split columns, remove duplicates	
5	Data Modeling & Relationships	Star schema basics; Model view; cardinality; Manage Relationships	2 hours
6	Introduction to DAX	Calculated columns vs. measures; SUM, AVERAGE, CALCULATE	
7	Creating Visualizations	Bar/column, line, pie, card, table, map visuals; formatting	2 hours
8	Designing Interactive Reports & Dashboards	Slicers, filters, drill-through, bookmarks, layout design	
9	Publishing & Sharing via Power BI Service	Publish, workspaces, dashboards, sharing, refresh	2 hours
10	Final Project & Assessment	Build an end-to-end report from a real dataset; presentation	

Total course duration: approximately 10 hours.

3. Detailed Training Notes

Module 1: Introduction to Business Intelligence & Power BI

1.1 What Is Business Intelligence (BI)?

Business Intelligence refers to the technologies and processes used to collect, clean, and analyze raw data and turn it into meaningful information that supports decision-making. Instead of scrolling through thousands of spreadsheet rows, a manager can look at a single dashboard and immediately see which products are selling, which regions are underperforming, or which program indicators are off track.

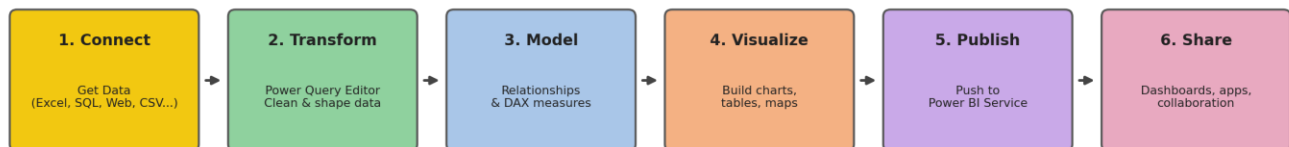
1.2 What Is Power BI?

Power BI is Microsoft's business intelligence suite for connecting to data, transforming it, modelling relationships, and presenting insights through interactive visuals, reports, and dashboards. It is widely used because it is free to start with, integrates tightly with Excel, and requires no programming knowledge to produce professional results.

1.3 The Three Building Blocks of Power BI

- Power BI Desktop: a free Windows application used to connect to data, transform it, build data models, and design reports. This is where most of this course takes place.
- Power BI Service (app.powerbi.com): the cloud platform used to publish, share, and collaborate on reports and dashboards.
- Power BI Mobile: apps for iOS and Android that let you view dashboards and reports on the go.

The Power BI Workflow



The end-to-end Power BI workflow: Connect → Transform → Model → Visualize → Publish → Share.

1.4 Why Learn Power BI?

- High demand across finance, NGOs/M&E, government, health, education, and private-sector reporting roles.
- Free entry point (Power BI Desktop) with a very active global community and official documentation.
- Directly builds on Excel skills most learners already have.

Tutor Tip

Open real dashboards (e.g., a national statistics dashboard or a sample sales dashboard) at the start of the very first class so learners can see the destination before they learn the steps.

Practice Exercise 1

1. In your own words, write two sentences explaining the difference between raw data and business intelligence.
2. List three real-world situations at your workplace, school, or organization where a dashboard would help decision-making.

Module 2: Installing Power BI Desktop & Touring the Interface

2.1 Installing Power BI Desktop

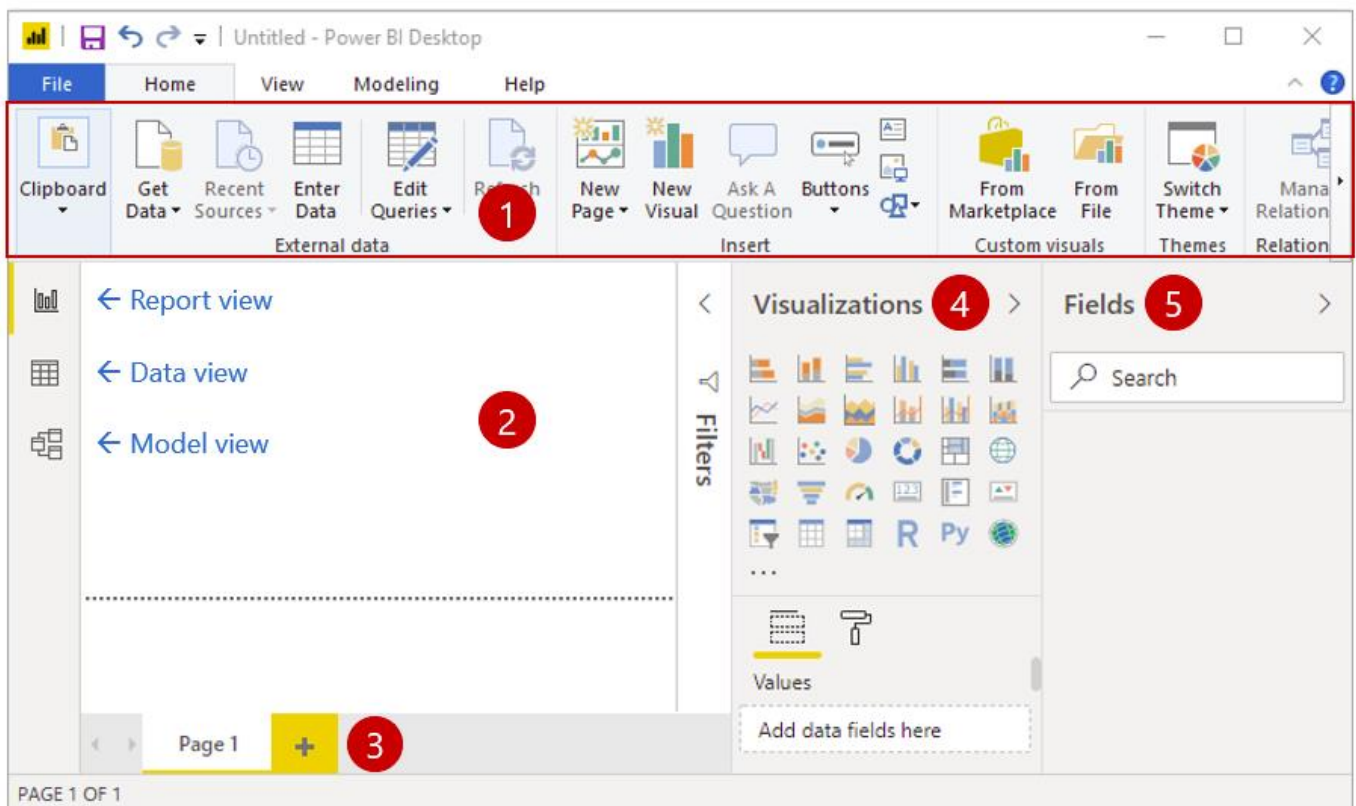
Power BI Desktop can be installed in either of two ways:

1. Recommended: Install from the Microsoft Store, which updates automatically.
2. Alternative: Download the installer directly from the Microsoft Download Center and run it.

Minimum system requirements include Windows 10 or later, at least 2 GB of available RAM (4 GB recommended), and a screen resolution of at least 1440 × 900.

2.2 Illustration — The Power BI Desktop Interface

The diagram below is a labelled illustration of the Power BI Desktop window. Learners should keep this page open on a second screen (or printed) during their first few practice sessions until the layout becomes second nature.



Labelled illustration of the Power BI Desktop interface.

2.3 Interface Walkthrough

- ① **The Ribbon (Home tab):** Displays common tasks that are associated with reports and visualizations.
- ② **The Report Canvas:** The large white working area in the center where visuals are placed, resized, and arranged. Each report can contain multiple pages, shown as tabs along the bottom.
- ③ **Pages tab:** Located along the bottom of the page, this area is where you would select or add a report page.
- ④ **The Visualizations Pane:** A gallery of chart-type icons (bar, line, pie, card, table, map, and more). Selecting an icon adds that visual to the canvas; the same pane also holds formatting options for the selected visual.
- ⑤ **The Fields Pane:** Lists every table and column available in your data model. Fields are dragged from here onto a visual, or added by ticking their checkboxes.
- ⑥ **The Status Bar:** Shows refresh progress, zoom controls, and the active page.

View switcher (left edge) three icons let you switch between Report view (build visuals), Data view (see raw table rows), and Model view (see and edit relationships between tables).

2.4 First Launch Walkthrough

- 1) Open Power BI Desktop from the Start menu.
- 2) Close or explore the welcome/tutorial screen that appears on first launch.
- 3) Note the blank Report canvas this is your empty starting point.
- 4) Hover over each ribbon tab (Home, Insert, Modeling, View) to preview available commands without clicking.
- 5) Click the three view icons on the left edge (Report, Data, Model) to see how the window changes return to Report view when done.

Tutor Tip

Ask learners to point to each pane on their own screen and name it out loud before moving on this quick recall check prevents confusion in later modules.

Practice Exercise 2

1. Install Power BI Desktop on your machine (or confirm it is already installed).
2. Open the application and, without clicking anything else, identify and label all six interface areas on your own screen.

Module 3: Connecting to Data Sources

3.1 The "Get Data" Command

All Power BI projects begin with connecting to data. From the Home ribbon, selecting Get Data opens a window listing hundreds of possible data sources, grouped into categories such as File, Database, Power Platform, Azure, and Online Services.

3.2 Common Beginner Data Sources

- Excel Workbook; the most common starting point; Power BI reads worksheets and Excel Tables.
- CSV / Text File; simple comma-separated files, useful for exported system data.
- Web; connect directly to a table published on a web page using a URL.
- SQL Server Database; for organizations with a structured database backend.

3.3 Step-by-Step: Importing an Excel File

1. On the Home ribbon, select Get Data > Excel workbook.
2. Browse to the file on your computer and select Open.
3. In the Navigator window, tick the checkbox next to each worksheet or table you want to import; a preview appears on the right.
4. Select Transform Data if the data needs cleaning first (recommended), or Load to bring it straight into the model.

3.4 The Navigator Window

The Navigator window is where you choose exactly which sheets or tables to bring into your model. Selecting a table shows a live preview of its rows and columns, helping you avoid loading unnecessary data.

Tutor Tip

Always prefer Excel Tables (Ctrl+T) over plain worksheet ranges when preparing source files; tables have defined boundaries and give Power BI cleaner, more reliable results.

Practice Exercise 3

1. Prepare a simple Excel file with one sheet of sample data (e.g., Product, Region, Sales Amount).
2. Import it into a new Power BI Desktop file using Get Data > Excel workbook.
3. Confirm the correct table appears in the Fields pane after loading.

Module 4: Cleaning Data with Power Query Editor

4.1 What Is Power Query Editor?

Power Query Editor is Power BI's built-in data-cleaning workspace. It opens when you select Transform Data from the Home ribbon, and every cleaning step you apply is recorded in the Applied Steps list on the right, so the process is fully repeatable and reversible.

4.2 Common Cleaning Operations

- Remove Columns; right-click any unwanted column header and choose Remove.
- Change Data Type; click the small type icon in a column header to switch between Text, Whole Number, Decimal, Date, and other types.
- Remove Duplicates; right-click a column and choose Remove Duplicates to eliminate repeated rows.
- Split Column; use Split Column (by Delimiter or by Number of Characters) to break combined fields such as "City, Country" into two columns.
- Filter Rows; use the drop-down arrow in any column header to remove blank rows or unwanted categories.
- Rename Columns; double-click a header to give it a clearer, business-friendly name.

4.3 Applied Steps & Close & Apply

Every action taken in Power Query Editor is listed as a step under Applied Steps. This list can be edited, reordered, or deleted, making it easy to correct mistakes. Once cleaning is complete, selecting Close & Apply saves the changes and loads the cleaned data back into the main Power BI window.

Tutor Tip

Encourage learners to rename each Applied Step (right-click > Rename) with a short description, e.g. "Removed blank rows" — this makes the query self-documenting for anyone who opens the file later.

Practice Exercise 4

1. Open the Excel file imported in Module 3 in Power Query Editor.
2. Remove at least one unnecessary column.
3. Correct the data type of at least one column (e.g., set a date column to Date type).
4. Select Close & Apply and confirm the cleaned data loads without error.

Module 5: Data Modeling & Relationships

5.1 Why Modeling Matters

Real-world data usually lives in more than one table; for example, a Sales table and a separate Products table. Data modeling is the process of telling Power BI how those tables relate to one another, so that a single report can pull correct, combined information from all of them.

5.2 The Star Schema (Simplified)

Beginners are introduced to the star schema idea: one or more central "fact" tables holding transactions (e.g., Sales) surrounded by "dimension" tables holding descriptive details (e.g., Products, Regions, Dates). This structure keeps reports fast and easy to understand.

5.3 Working in Model View

- Switch to Model view using the view switcher on the left edge of the window.
- Tables appear as boxes; dragging a field from one table onto a matching field in another table creates a relationship, shown as a connecting line.
- Double-clicking a relationship line opens Edit Relationship, where you can check cardinality (one-to-many, many-to-many) and cross-filter direction.

5.4 Manage Relationships

The Manage Relationships button (Modeling ribbon) lists every relationship in the model, letting you review, edit, or delete connections without hunting through the diagram.

Tutor Tip

Use a simple two-table example (Sales and Products linked by Product ID) for the first hands-on relationship exercise before introducing more complex models.

Practice Exercise 5

1. Import a second small table (e.g., a Products list) alongside your Module 3 data.
2. Switch to Model view and create a relationship between the two tables.
3. Open Manage Relationships and confirm the new relationship is listed correctly.

Module 6: Introduction to DAX

6.1 What Is DAX?

DAX (Data Analysis Expressions) is the formula language used throughout Power BI to build calculated columns and measures. Learners who know basic Excel formulas will find many DAX functions familiar.

6.2 Calculated Columns vs. Measures

- Calculated Column; a new column added to a table, calculated row by row (e.g., Profit = Sales[Revenue] - Sales[Cost]).
- Measure; a dynamic calculation evaluated in the context of whatever filters or visuals are currently applied (e.g., Total Sales = SUM(Sales[Amount])). Measures are the preferred approach for most report calculations because they are efficient and context-aware.

6.3 Common Beginner DAX Functions

- SUM() — adds all values in a column.
- AVERAGE() — calculates the mean of a column.
- COUNT() / COUNTROWS() — counts values or rows.
- CALCULATE() — changes the filter context of another calculation, e.g., Total Sales 2025 = CALCULATE([Total Sales], Dates[Year] = 2025).

6.4 Creating a Measure

- 1) On the Home or Modeling ribbon, select New Measure.
- 2) Type a formula in the formula bar, e.g., Total Sales = SUM(Sales[Amount]).
- 3) Press Enter — the new measure appears in the Fields pane and can be dragged onto any visual.

Tutor Tip

Keep the first DAX session to just SUM and AVERAGE measures — save CALCULATE and time-intelligence functions for a follow-on intermediate course.

Practice Exercise 6

1. Create a measure that sums your sales/amount column.
2. Create a second measure that averages the same column.
3. Add both measures to a Card visual to see the results update live.

Module 7: Creating Visualizations

7.1 Choosing the Right Visual

- Bar / Column Chart; best for comparing categories (e.g., sales by region).
- Line Chart; best for showing trends over time.
- Pie / Donut Chart; best for showing proportions of a whole, with only a few categories.
- Card; best for displaying a single key number (e.g., Total Sales).
- Table / Matrix; best for detailed, row-level data or cross-tabulated summaries.
- Map; best for geographic data such as sales by country or region.

7.2 Building a Visual

- 1) In the Visualizations pane, click the icon for the chart type you want; an empty placeholder appears on the canvas.
- 2) Drag fields from the Fields pane into the Axis, Legend, and Values boxes shown in the Visualizations pane.
- 3) Resize and reposition the visual by dragging its corners and edges on the canvas.
- 4) Use the Format (paint-roller) tab in the Visualizations pane to adjust colours, titles, data labels, and axis settings.

7.3 Formatting Best Practices

- Give every visual a clear, descriptive title.
- Use consistent colours for the same category across all visuals on a page.
- Avoid overcrowding a page — three to six well-chosen visuals communicate more than twelve cramped ones.

Tutor Tip

Have learners build the same three-chart layout shown in Figure 2.1 (bar, pie, line) as their first guided exercise, since it mirrors the illustration, they already studied in Module 2.

Practice Exercise 7

1. Create a bar chart comparing a category field against a numeric measure.
2. Create a line chart showing the same measure over a date field.
3. Add titles to both visuals and apply a consistent colour theme.

Module 8: Designing Interactive Reports & Dashboards**8.1 Slicers and Filters**

A Slicer is a visual filter placed directly on the report page, letting viewers pick a category, date range, or value and instantly see every other visual on the page update. Slicers make a report interactive rather than static.

8.2 Drill-Through and Bookmarks

- Drill-through; lets a viewer right-click a data point on one page and jump to a detail page filtered to that specific item.
- Bookmarks; capture the current state of a report page (filters, visibility, zoom) so it can be recalled instantly, useful for guided "story" navigation through a report.

8.3 Layout & Design Principles

- Place the most important number or chart in the top-left corner, where the eye naturally lands first.
- Group related visuals together and leave breathing space between elements.
- Use a consistent theme (View > Themes) so colors and fonts match across the whole report.

8.4 From Report to Dashboard

A Report is a multi-page, highly interactive Power BI file. A Dashboard (built later in the Power BI Service) is a single page of pinned visual "tiles" pulled from one or more reports, designed for a quick, at-a-glance overview.

Tutor Tip

Introduce one slicer per page maximum in the first exercise — too many filter options can overwhelm beginners before they understand the underlying concept.

Practice Exercise 8

1. Add a slicer to your report page based on a category field (e.g., Region or Product).
2. Test the slicer by selecting different values and observing how the other visuals update.
3. Apply a built-in theme to the whole report from the View ribbon.

Module 9: Publishing & Sharing via Power BI Service

9.1 What Is the Power BI Service?

The Power BI Service (app.powerbi.com) is the cloud home for Power BI content. Publishing a report from Desktop uploads it to a workspace in the Service, where it can be viewed in a browser, shared with colleagues, and combined into dashboards.

9.2 Publishing a Report

1. In Power BI Desktop, select Publish from the Home ribbon.
2. Sign in with a Power BI account if prompted.
3. Choose a destination workspace (e.g., My Workspace for personal practice).
4. Wait for the upload to complete, then select the link to open the report in the browser.

9.3 Sharing and Dashboards

- Reports can be shared directly with named colleagues or via a shareable link, depending on licensing.
- Pinning a visual from a report creates a tile on a Dashboard, which can combine tiles from several different reports onto one summary page.
- A Power BI Pro or Premium Per User license is generally required to share content with others in an organization; My Workspace can be used freely for personal practice.

9.4 Keeping Data Fresh

Scheduled Refresh, configured in the workspace settings, keeps a published report's data up to date automatically, so viewers always see current figures without the report author manually republishing the file.

Tutor Tip

If learners do not yet have organizational Power BI accounts, demonstrate publishing using a trial account so the class still sees the full Desktop-to-Service workflow.

Practice Exercise 9

1. Publish your practice report to My Workspace in the Power BI Service.
2. Open the published report in a web browser and confirm it matches the Desktop version.
3. Pin one visual to a new dashboard.

Module 10: Final Project & Assessment

10.1 Capstone Project Brief

Working individually or in small groups, learners select a small real or sample dataset (for example, school enrolment figures, a shop's sales records, or a community health indicator set) and build a complete, end-to-end Power BI report.

10.2 Project Requirements

1. Connect to and clean the chosen dataset using Power Query Editor.
2. Build a simple data model with at least one table relationship (if more than one table is used).
3. Create at least one DAX measure.
4. Design a report page with at least three different visual types.
5. Add at least one slicer to make the report interactive.
6. Publish the finished report to the Power BI Service (or export as a .pbix file if no account is available).

10.3 Assessment Rubric (Summary)

- Data preparation and cleanliness 20%
- Correct use of relationships and DAX 20%
- Choice and clarity of visuals 25%
- Interactivity (slicers/filters) 15%
- Overall design and presentation 20%

10.4 Wrap-Up

Each learner presents their dashboard in a short 3–5 Minute walkthrough, explaining the story their data tells and the choices they made. Certificates of Completion are awarded to learners who meet the project requirements.

4. Glossary of Key Terms

- Canvas: the working area of a report page where visuals are placed.
- DAX (Data Analysis Expressions): the formula language used for measures and calculated columns.
- Dashboard: a single page of pinned tiles from one or more reports, built in the Power BI Service.
- Dataset / Semantic Model: the collection of tables, relationships, and measures behind a report.
- Measure: a dynamic DAX calculation that responds to filters and context.
- Power Query Editor: the tool used to connect, clean, and reshape data before it is loaded into the model.
- Relationship: a defined connection between two tables based on a shared field.
- Ribbon: the toolbar of commands at the top of Power BI Desktop, organised into tabs.
- Slicer: an on-page visual filter that lets viewers interactively filter a report.
- Visual: any chart, table, map, or card placed on the report canvas.

5. References

The following official Microsoft Learn resources were used to prepare these notes and are recommended for further independent study:

<https://learn.microsoft.com/en-us/power-bi/>

<https://learn.microsoft.com/en-us/power-bi/fundamentals/>

<https://learn.microsoft.com/en-us/power-bi/fundamentals/desktop-getting-started>

<https://learn.microsoft.com/en-us/power-bi/fundamentals/desktop-get-the-desktop>

<https://learn.microsoft.com/en-us/training/modules/get-data-power-bi/2-getting-started-power-bi-desktop>

<https://learn.microsoft.com/en-us/power-bi/fundamentals/service-get-started>

<https://learn.microsoft.com/en-us/training/modules/get-started-with-power-bi/>

<https://www.microsoft.com/en-us/power-platform/products/power-bi/desktop>

— *End of Lesson Guide* —

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